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Digital Safety Navigation for Tower Crane Climbing Process from a Systematic Perspective: Risk Identification, Safety Information Requirements, and Related Intelligent Platform

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Abstract: The tower crane climbing process is one of the most hazardous operations due to its complexity and technical challenges, making it difficult to ensure both structural and worker safety. To address this issue, this study proposes a systematic control framework based on digital navigation. This framework integrates theoretical risk analysis, safety information requirements, and the practical implementation of a safety intelligent navigation platform (SINP). Key safety-critical points in the tower crane climbing process are identified through process analysis, and worker situation awareness requirements are derived using goal-directed task analysis (GDTA). The SINP is designed to digitally capture safety-related data, assess safety conditions intelligently, and provide automated safety guidance. The platform enables systematic control of tower crane safety through three key functions: (1) real-time navigation and alarms for workers via an edge intelligent control device (EICD); (2) synchronous monitoring and interaction for on-site supervisors using mobile devices; and (3) remote inspections for management through cloud-based platforms. Two experimental scenarios were conducted to simulate real-world engineering conditions and evaluate the framework's effectiveness. Results show a significant improvement in worker performance and process safety, which are attributed to the advantages of SINP, which include (1) timely and efficient information sharing; (2) accurate and orderly safety guidance; (3) real-time and effective alarms; and (4) positive psychological reinforcement for workers. This study introduces a proactive safety management approach that can prevent accidents at their source and enhance worker performance throughout the tower crane climbing process.
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Author keywords: Construction engineering; Tower crane climbing; Safety navigation; Information requirements; Digital twin; Edge intelligence.

Introduction

The construction industry serves as the backbone of the national economy and plays a critical role in the long-term development of both society and the economy. In 2022, the industry's gross domestic product (GDP) reached 8.3 trillion yuan, contributing 6.9% to China's overall GDP, marking a 5.9% increase compared to 2021 (National Bureau of Statistics 2022). The workforce in this sector totaled 18.35 million, representing 11.0% of the country's total labor force. By the end of 2022, the number of special equipment units had reached 19.55 million, including 2.79 million units of lifting machinery, accounting for 14.3% (Chinese SMAR 2023). As shown in Fig. 1, 108 safety incidents related to special equipment were reported in 2022, with 25 involving cranes, representing 23.1% of the total. These crane-related accidents led to 101 fatalities, of which 25 (22.8%) were directly linked to crane operations, predominantly in the construction sector. These statistics underscore the persistent and serious safety challenges associated with crane operations in the construction industry.

Tower cranes are widely used in construction for lifting materials due to their adjustable height and powerful lifting capacity (Wu et al. 2020). However, accidents involving tower cranes remain frequent and severe due to their unstable structural characteristics and the inherent risks of their operational tasks. Notably, the installation (including assembly and climbing) and dismantling of tower cranes are particularly high-risk activities, often leading to fatalities (Zhang and Zhang 2019). According to the Occupational Safety and Health Administration (OSHA), crane accidents resulted in over 400 nonfatal injuries in 2020, with an average of 33 fatalities annually over the past decade (US BLS 2023). In Korea, 245 fatal tower crane accidents were reported between 2007 and 2016, with human error and crane malfunctions identified as the primary causes (Lee et al. 2022). Additionally, incomplete statistics on tower crane accidents in China from 2010 to 2019 indicate that 39.8% of the 344 reported incidents occurred during installation or dismantling, with 47 of these taking place during the climbing process. As construction

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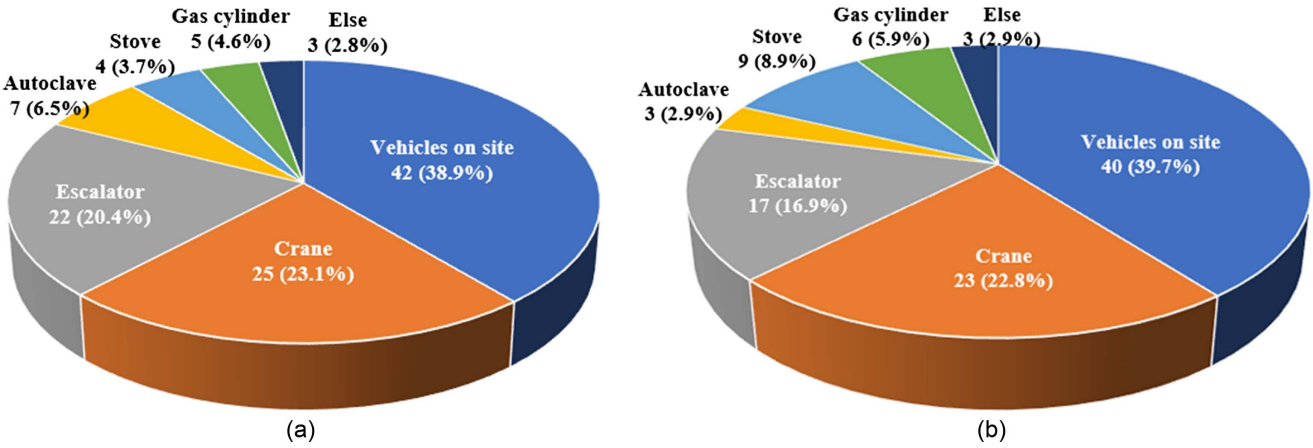


Fig. 1. Accidents of special equipment in 2022: (a) number of occurrences; and (b) number of deaths.

efficiency continues to improve, the frequency of crane climbing operations is expected to increase, further elevating safety concerns. Therefore, enhancing the safety of tower crane climbing operations is becoming an increasingly critical focus.

The growing demand for tower cranes in construction, along with frequent accidents, has highlighted the critical need for effective safety management during installation and dismantling, particularly in the climbing process. In China, government departments across various provinces and cities, such as Hubei (HP 2023), Chongqing (CQ 2021), and Guangxi (ZJT 2021), have implemented policies to enhance intelligent monitoring and safety management for tower cranes. For example, the Department of Housing and Urban-Rural Development of Hubei Province issued an emergency notice emphasizing the need for safety monitoring devices to ensure standardized operations during crane climbing and dismantling (HP 2022). This policy was quickly adopted by related departments in Xianning, Xiaogan, and Yichang.

Revolutionary technologies and tools are urgently needed to address key safety challenges in the tower crane climbing process. These challenges include:

Time-space separation between different working planes. This separation makes it difficult for workers on different planes to communicate with each other and leads to the operation scene being out of supervisors' vision, which makes it hard for workers and supervisors to perceive the real-time status of other planes and make decisions in time, as shown in Fig. 2.

- Poor information of risks influencing the process safety. Workers may execute unreasonable operations based on biased comprehension of tower crane status because of the lack of critical information of risks.
- Limited cognitive awareness of safety protocols among operation staff. Workers usually irrationally prioritize efficiency over safety due to insufficient awareness of potential accidents, leading to compromised operation quality and deviations from prescribed procedures.

Given the limitations of existing methods, which primarily focus on structural health and environmental monitoring, it is important to recognize that tower cranes are highly prone to collapse, leaving minimal time to react to emergencies once abnormalities are detected (Jiang et al. 2022). Instead of relying on passive monitoring and reactive responses, this study proposes a systematic control framework that proactively guides operations and delivers timely warnings of unsafe conditions during the climbing process. The goal of this framework is to enhance worker safety, prevent accidents from the outset, and ensure the overall safety of the climbing operation. It is built around three key components:

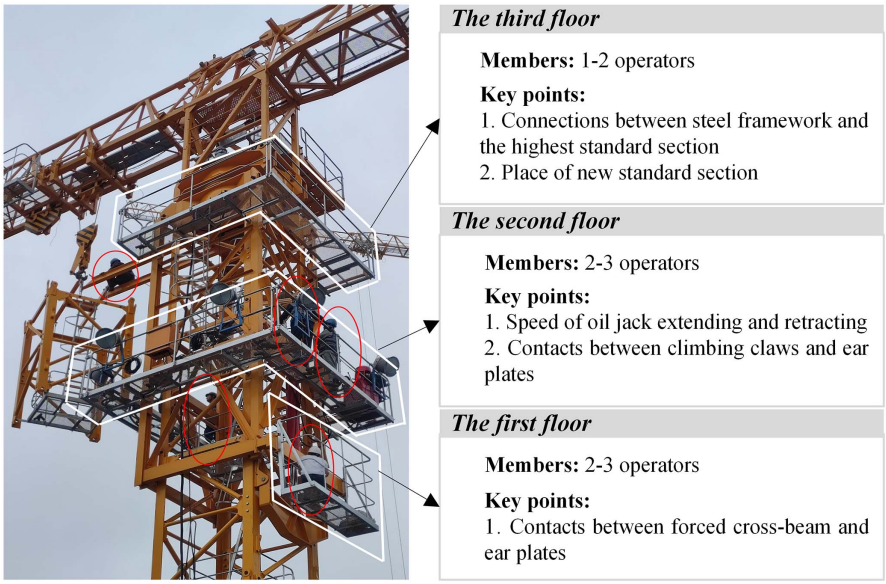


Fig. 2. Workers in different working planes during tower crane climbing. (Images by Zhang Chong.)

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1. Identification of risks during the climbing process: Regular operations and satisfied performance could be clarified, and safety boundaries for each step are established.
2. Determination of safety information requirements: This creates a basis for implementing safety navigation and real-time alarms.
3. Development of the safety intelligent navigation platform (SINP): This platform integrates digital and intelligent technologies to enhance operational performance and ensure of safety of the climbing process.

This study is structured according to the proposed systematic control framework. Following a brief introduction to the topic and identification of research gaps, the related works are discussed in detail. Next, a comprehensive description of the framework is provided, along with an analysis of the risks and safety information requirements. The development of the safety intelligent navigation system (SINS) is then presented, followed by a performance evaluation through two experimental scenarios. Finally, the key contributions and limitations of the study are summarized.

Related Work

Tower Crane–Related Safety Management

Tower cranes play a crucial role in construction projects, serving as indispensable tools for lifting building materials. Research on tower crane safety has spanned decades, primarily focusing on two key areas: the identification and assessment of safety risks and the implementation of monitoring measures to ensure safety.

Two primary methods are commonly used to identify and assess tower crane risks: statistical analysis and interview studies. Through analyzing large volumes of accident data, researchers have found that complex procedures and poor worker performance are the main contributors to high risks, particularly during the installation phase (Aneziris et al. 2008; Beavers et al. 2006; Swuste 2013). This conclusion is supported by several studies on tower crane collapses, where worker errors in understanding crane status and executing operations were frequent causes of failure (Frendo 2013; Marquez et al. 2014; McDonald et al. 2011; Zrnic et al. 2011). Additionally, a quantitative analysis of 212 incidents confirmed that the most hazardous factors are technical and human errors (Raviv et al. 2017a, b). In fact, 148 operator errors were identified, with risks quantified accordingly (Borgheipour et al. 2020). Interviews and surveys have further highlighted and ranked the factors influencing tower crane safety. These include inadequate worker training, complex procedures, deteriorating equipment, insufficient supervision, and poor working conditions (Kim and Singhose 2010; Sertyesilisik et al. 2010; Sharma et al. 2019). It is widely acknowledged that these issues play a major role in incidents occurring during installation and dismantling (Shin 2015; Tam and Fung 2011). Moreover, several studies have explored broader factors contributing to tower crane accidents, with findings largely consistent with earlier research (Zhang et al. 2020, 2021; Zhou et al. 2018).

In terms of monitoring measures, passive monitoring is widely used to gather information on the status of hooks, lifting capacity, and limiting devices, with extensions to static monitoring of structural safety indicators such as stress, displacement, and wind speed. Condition monitoring systems provide detailed real-time data on factors like lifting moment, swing amplitude, and lifting weight, using sensing, signal processing, and wireless technologies (Zhang et al. 2013). Sensor data, including lifting weight, boom inclination, rotation angle, and wind speed, are captured and used in real-time FEM analysis to assess the structural safety of the tower crane (Guanghui et al. 2020). Key parameters monitored for structural safety include stress, deformation, temperature, wind load, and acceleration (Chong et al. 2021; Zhang et al. 2021). Structural stress, verticality, and wind speed are compared against predefined thresholds to determine warning levels (Wei et al. 2021). By monitoring hook height, trolley position, and jib angle, visual monitoring systems can analyze driver behavior through a scoring mechanism to improve operational safety (Yannian et al. 2022). Additionally, this monitoring data can be utilized for model analysis and simulation of crane operations and accidents. The safety level of tower cranes at individual sites can be predicted using an integrative model that analyzes the impact of various overall safety indices (Shapira et al. 2012). Additionally, a multiuser virtual safety training system has been developed to simulate real-world conditions for workers involved in tower crane dismantling, offering a practical and immersive learning environment (Li et al. 2012).

Machine learning (ML) has been applied to safety hazard identification, error detection, and the recognition of unsafe behaviors. A smart packaging method was proposed to enable adaptive decision-making and path replanning in dynamic crane operation environments (Li et al. 2020). An audio-based safety recognition system was also developed to continuously monitor and manage safety risks by detecting abnormal events (Lee et al. 2020). Additionally, a deep learning algorithm has been used to predict operator fatigue during crane operations (Liu et al. 2021). Further studies have expanded these efforts (Jiang and Ding 2024; Wu et al. 2022). With advancements in computing and storage capabilities, there has been increasing integration of traditional methods with emerging technologies such as artificial intelligence (AI), blockchain, and digital twin (DT). A blockchain-based framework was developed to enhance the effectiveness of on-site inspections of tower cranes (Wu et al. 2022). Using DT technology, a general framework was proposed to analyze unsafe behaviors and improve human-machine interactions in construction activities (Jiang et al. 2022). Moreover, a DT-driven model was created to predict the reduced lifting capacity of aging tower cranes using real-time data, thereby enhancing safety (Hussain et al. 2024).

In summary, extensive research has been conducted on the identification and assessment of safety risks in tower crane operations. It is widely acknowledged that complex procedures and inadequate execution are the two most significant factors contributing to tower crane incidents, particularly during installation and dismantling. Most existing safety management approaches rely on passive monitoring and issuing warnings, which often prove ineffective in emergency situations. While emerging technologies have been explored to improve safety, most of these efforts remain in the experimental phase and have not yet been implemented in practical projects. Furthermore, there is still no efficient solution for managing the safety risks associated with complex procedures and inadequate operations during the installation process.

Information System for Construction Safety Management

Information systems are widely used to enhance construction safety management across various areas, including safety in design (Zhang and Hu 2011), site safety planning (Cheng and Teizer 2013), safety information management (Shi and Xu 2021), safety knowledge databases (Zhang et al. 2015), safety monitoring (Jiang 2020), and safety training (Park and Kim 2013). Among these, safety information management systems are the most common, utilized by project departments, enterprises, and government agencies. Detailed records of daily operations and accident-related events are crucial for addressing on-site accidents (de la Garza et al. 1998; Hinze and Russell 1995). For instance, accident data is recorded and stored

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within the system, providing a comprehensive basis for assessing safety performance (Skibniewski 2014). A decentralized accident information management system has also been proposed to identify root causes and prevent future incidents based on collected data (Ahmadisheykhsarmast et al. 2023). Additionally, safety monitoring has advanced with the development of information technologies. A cloud-based monitoring application has been developed to track environmental conditions and workers' health status, using real-time data to detect emergencies (Wu et al. 2019). Furthermore, sensor-based systems represent a promising direction for safety information analysis and management (Skibniewski 2014).

In summary, information systems play a critical role in construction safety management and have been successfully implemented in numerous projects, with continual improvements over time. However, limited research exists on information systems specifically designed for tower crane installation, particularly regarding the safety management of complex procedures and operations.

Goal-Directed Task Analysis for Safety Information

Eliciting end users' information requirements is a critical aspect of the information system (IS) development cycle, aiming to identify the data users need to make informed decisions. Goal-oriented analysis has been proposed in requirements engineering to capture the intent behind software requirements, providing an intuitive approach to eliciting and analyzing user needs. Among these methods, goal-directed task analysis (GDTA) was developed to enhance situation awareness (SA) for end users, particularly in aviation and military domains, and has been widely adopted for its efficiency and ease of use, as shown in Fig. 3.

GDTA is a cognitive task analysis method that provides a detailed description of the SA requirements of human operators within a system (Endsley et al. 1998; Endsley 2001). It is an effective tool for identifying the necessary information operators need to understand the current situation and make decisions to achieve their task goals (Bolstad et al. 2002). Goal-directed information analysis, based on GDTA, has been proposed as a protocol to capture the specific information needs in high-risk and emergency-related domains (Prasanna 2009). This method has been successfully applied in industries such as maritime navigation (Cordon et al. 2017) and firefighting (Bearman and Bremner 2013), both of which involve planned procedures, unpredictable events, and significant risks. For instance, GDTA was used to define goals and information requirements for maritime pilotage, contributing to the design of operating procedures and improving the understanding of maritime accidents (Sharma et al. 2019). Similarly, it was applied to fire emergency response systems to improve decision-making, especially in situations where firefighters lacked the skills and training needed to manage emergencies (Rezaeifam et al. 2023). In aviation, GDTA has been combined with FCM (fuzzy cognitive maps) to establish situation awareness models for flight crews, ensuring sufficient SA during operations (Li 2022; Lijing et al. 2022). A modified GDTA method has also been used in goal–decision–process interviews to clarify submariners' SA information requirements during sampling tasks (Wang et al. 2024). Moreover, GDTA has been employed to identify key goals (e.g., work process, quality, cost, safety, and communication), subgoals, decisions, and the necessary information, helping to reveal critical requirements for future developments and illustrating the relationships between goals and information (Pillajo et al. 2023).

In tower crane installations, while workers are provided with prescheduled procedures and standard operation instructions, the specific information requirements for various scenarios are often unclear, leading to errors. GDTA is a highly suitable and effective tool for addressing this issue.

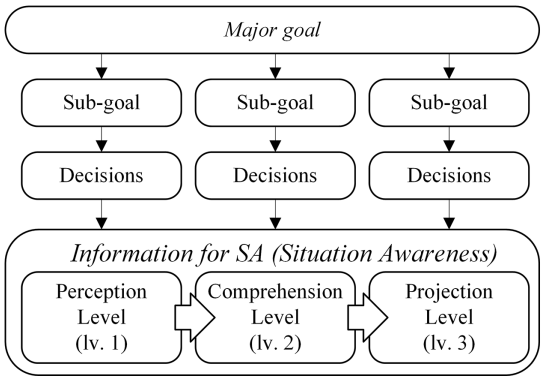


Fig. 3. Goal–decision–information structure of SA.

Methodology

The proposed systematic control framework consists of three key components: risk identification for process safety, information requirements for risk control, and the development of a safety intelligent navigation platform. As illustrated in Fig. 4, the first component identifies potential risks arising from complex procedures and inadequate operations. The second component defines the information requirements for risk control through a hierarchical decision-making process analysis. The final component involves building a safety intelligent navigation platform that integrates a physical model, virtual model, real-time database, and service core. This platform creates a closed-loop system that provides safety navigation and real-time alarms to ensure the safety of the tower crane climbing process.

Risk Identification and Safety Information Requirements

To ensure the safety of the tower crane climbing process, it is essential to identify the factors that influence safety and determine the necessary information for risk control. The structure of a tower crane is dynamic and unstable during the climbing process, with the internal forces between the steel framework and the tower constantly shifting. This requires workers to perform precise operations and maintain crane stability at all times. A detailed analysis of the climbing process is conducted to identify risks that could lead to hazardous situations, with operational guidelines clearly defined for managing these risks at each stage. Based on the identified risks, a GDTA is performed to determine the safety information requirements for risk control. This information is crucial for workers to understand the quality of their operations and the current safety status of the tower crane, allowing them to respond effectively to inadequate operations and abnormal conditions.

Risk Identification for Process Safety

A complete tower crane climbing cycle consists of eight steps, which can be divided into two stages of steel framework climbing and one stage for deploying a new section, as shown in Fig. 5.

1. The weight of the steel framework is transmitted to the tower crane body through the support between the climbing claws and ear plates [see Fig. 6(a)], while the oil jack remains retracted.
2. The oil jack extends downward until the forced cross-beam rests on the ear plates [see Fig. 6(b)], shifting the load from the claws to the beam-plate.

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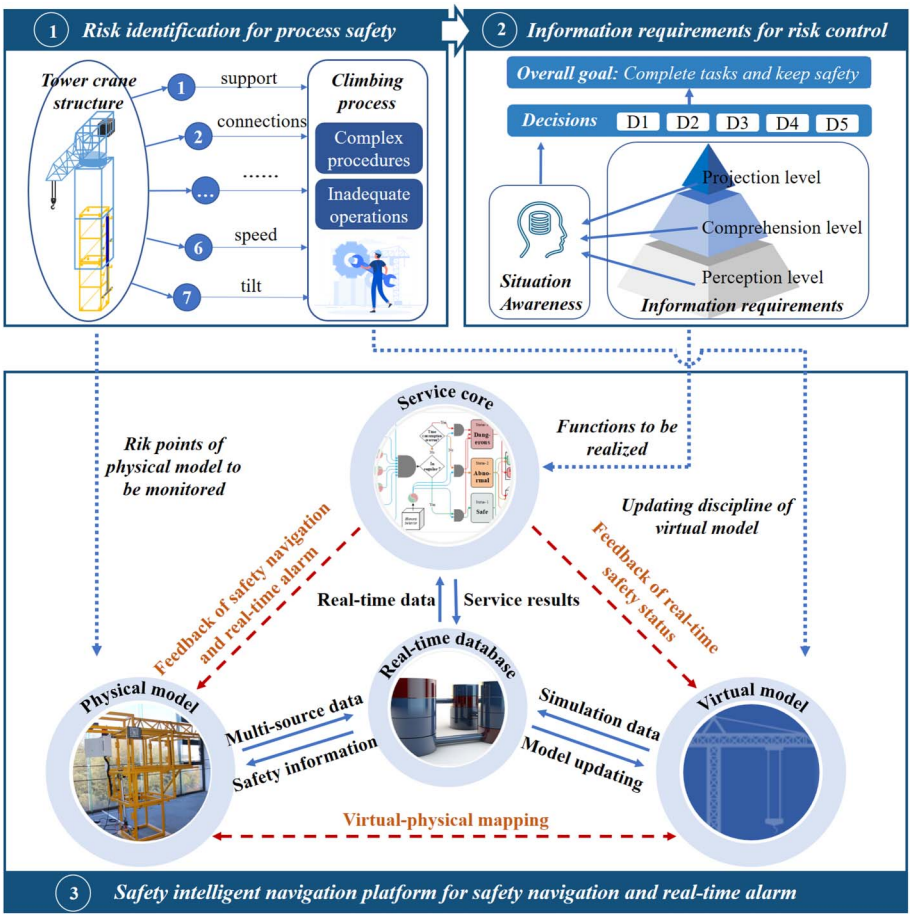


Fig. 4. Overall flowchart of the proposed systematic controlling framework.

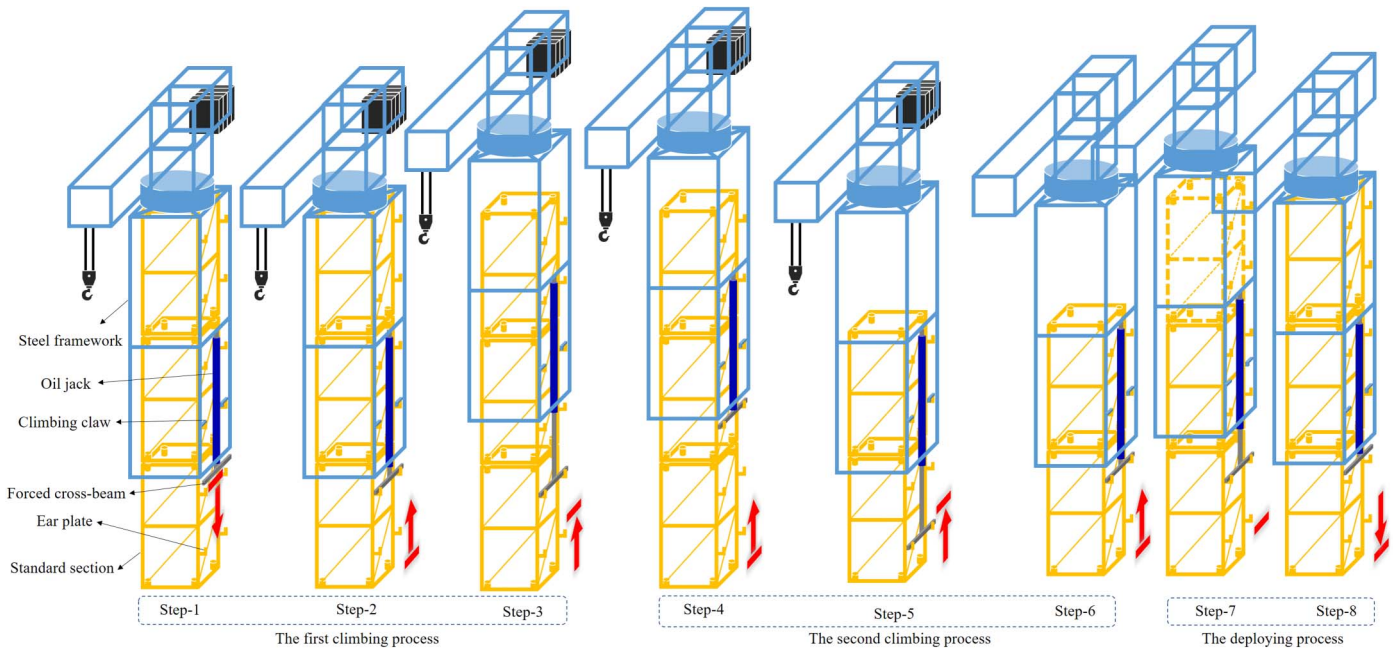


Fig. 5. Steps of a climbing cycle.

3. The oil jack extends upward until the climbing claws rest on the ear plates, reversing the load transfer from the beam-plate back to the claws. This completes the first stage of steel framework climbing.
4. The oil jack retracts, once again shifting the load to the forced cross-beam on the ear plates.
5. The oil jack extends upward, transferring the load back to the climbing claws.

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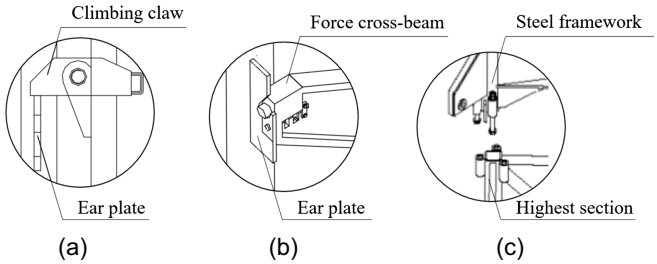


Fig. 6. Key structural parts: (a) support between climbing claw and ear plate; (b) support between force cross-beam and ear plate; and (c) connection between steel framework and the highest section.

6. The oil jack retracts, once again shifting the load to the cross-beam. This completes the second stage of steel framework climbing.
7. The oil jack extends upward, creating enough space to accommodate a new standard section.
8. The new standard section is then accurately positioned [see Fig. 6(c)]. The oil jack retracts downward, resting the forced cross-beam on the ear plates, and then retracts upward, returning the tower crane to its initial state, completing the new section deployment.

As a large-scale steel structure and specialized mechanical equipment, tower cranes have complex designs and inherent safety risks. During jacking operations, the crane undergoes multiple structural transitions, each requiring precise and tightly controlled operations to ensure the crane remains balanced and stable. This places high demands on both the overall process order and the quality of local operations. Maintaining a stable crane is not only the outcome of proper operations but also a prerequisite for the next phase of work. However, it is difficult to perceive the real-time safety status of the crane during the jacking process. The orderly execution of operations relies heavily on effective information exchange and collaboration among workers. However, workers often operate at high altitudes on multiple platforms, which severely hinders timely communication and coordination. The quality of local operations also depends on worker performance, but factors such as limited visibility, restricted working space, and varying levels of safety awareness make it challenging to consistently meet operational standards and accurately assess conditions. Additionally, the physical and temporal separation between safety supervisors and operators reduces the effectiveness of traditional safety supervision. These challenges pose significant threats to tower crane safety and must be addressed, as referenced in the Introduction section.

Given the constant dynamic changes in the crane's structural system, it is challenging to directly monitor and assess its overall safety status. The safety performance of the tower crane throughout the entire climbing cycle hinges on several critical risk points. The operational guidelines for controlling these risks at each step are outlined in Table 1.

Safety Information Requirements for Risk Control

To identify the safety information required during the tower crane climbing process, a series of interviews, theoretical analyses, and case studies were conducted using GDTA. Based structured and semistructured interviews, we summarized and validated the safety information requirements (see Tables 2 and 3) with various stakeholders, including operators, safety supervisors, manufacturers, and safety experts (see Table 4). The overall goal was defined, along with five categories of decisions (D1, D2, D3, D4, and D5) necessary to achieve it. For each decision, information was categorized into three levels: perception level (lv.1, e.g., I1.1.1), comprehension level (lv.2, e.g., I1.1), and projection level (lv.3, e.g., I1).

The overall goal of tower crane climbing is to complete the climbing task while ensuring occupational safety. Achieving this goal requires five basic safety decisions: (1) stop unsafe behaviors of operating staff, (2) check safety status and address safety hazards, (3) confirm initial conditions, (4) verify performance at each step, and (5) intervene in operations. Safety information is needed at three levels to support these decisions. To make safety decision D1, identify and stop unsafe behaviors of operating staff, it is necessary to assess their safety status (I1 at lv.3) based on the following factors: guidelines for safe behavior (I1.1 at lv.2), state of health (I1.2 at lv.2), and compliance with safety behaviors (I1.3 at lv.2). The comprehension of staff behavior compliance (I1.3 at lv.2) is further determined by the status of safety hardhats (I1.3.1 at lv.1), safety belts (I1.3.2 at lv.1), and physical behaviors (I1.3.3 at lv.1).

Throughout the entire tower crane climbing process, numerous decisions must be made at the right moments to ensure safe and orderly operations, as shown in Fig. 7. Decisions such as D1, D2, D3, D4, and D5 (Table 4) are crucial for achieving the overall safety and performance objectives. Initially, D2 is employed to establish a secure foundation for subsequent climbing tasks. At each step in Fig. 4, D3 is needed to confirm operational readiness, including personnel, crane, and environmental conditions. D4 ensures proper task execution, minimizing risks associated with inadequate operations. Additionally, D1 or D5 may be activated by specific safety risks, such as unsafe behaviors, operational errors, or hazardous conditions at critical points. These events disrupt the normal climbing process, requiring a corrective phase before operations can resume. After all major goals are completed, D2 must be executed to perform a comprehensive inspection of the tower crane, ensuring it is safe and stable before workers leave the site. To support the timely and effective execution of these decisions, a substantial amount of safety-related information must be efficiently gathered, transmitted, processed, and applied.

SINP Establishment for Safety Navigation and Real-Time Alarms

Based on the identified risks and safety information requirements, the SINP is designed to provide safety navigation and real-time alarms to enhance both operational efficiency and safety during

Table 1. Operation disciplines of key risk points in each step

Key risk points	S-1	S-2	S-3	S-4	S-5	S-6	S-7	S-8
Support between climbing claws and ear plates	Yes	No	Yes	No	Yes	No	No	Yes
Support between forced cross-beam and ear plates	No	Yes	No	Yes	No	Yes	Yes	No
Connections between steel framework and the highest standard section	Yes	No	No	No	No	No	No	Yes
Place of new standard section	No	No	No	No	No	No	No	Yes
Speed of oil jack extending and retracting	0 m/s				≤0.02 m/s			
The tilt of tower crane				≤4/1,000				
Wind speed				≤9 m/s				

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Table 2. Interview questions and answers for GDTA

Focus	Questions	Answers
Goals	What are the primary goals and subgoals of the tower crane climbing?	"Our primary goal is to elevate the tower crane to a specified height while ensuring personnel safety."
Decisions	What decisions must be made to achieve the goals?	"We need to conduct comprehensive safety inspections before commencing operations to eliminate potential safety risks." "We should assess the quality of work upon the completion of each operational phase to ensure compliance with standards." ...
	Are there any limitations in the process of making these key decisions?	"Sometimes, we may not be able to obtain timely information, leading to delayed responses." ...
Perception-level information	What direct information do you typically collect in real time during operations, on-site supervision, or remote monitoring?	"We instinctively observe the status of key components in the climbing mechanism, such as the climbing claws and jacking beams." "We want gather various information, including environmental conditions, structural safety indicators, the status of the workers, and specific progress." ...
	Have you encountered situations where certain important information is difficult to obtain while collecting this data?	"Information regarding the status of key components in blind spots is challenging to obtain." "The conditions of the high-altitude work platform cannot be directly observed." ...
Comprehension-level information	Why do you collect this direct information?	"This information allows us to understand the overall status of the tower crane's climbing mechanism and the progress of the operations." "It enables us to monitor the activities of the workers and the structural dynamics of the tower crane."
	What factors might affect your comprehension?	"My judgment may be influenced by stress and anxiety, which can impair my decision-making ability"
Projection-level information	What assistance does cognitive information provide you?	"This allows me to determine whether we can proceed to the next operation. Regarding relevant parameters." "This enables me to identify potential safety risks in advance." ...

the tower crane climbing process. The SINP is composed of four key components: a physical model, a virtual model, a real-time database, and a core service module. The physical model, equipped with various sensors, simulates practical operations and detects the safety risks identified during the process. The virtual model mirrors the climbing operations and structural status of the physical model. The real-time database serves as the engine facilitating interaction between the physical and virtual models. It stores real-time data from the physical model, which is used to update the status of the virtual model. Finally, the core service module provides safety navigation and real-time alarms, driven by the intelligent algorithms deployed on the platform.

Physical Tower Crane for Operation

The physical model is a fundamental component for risk monitoring and data acquisition, equipped with a variety of sensors and monitoring devices to capture real-time risk data, as shown in Fig. 8. This model replicates the power components and movable structures of a real tower crane, effectively simulating all climbing operations with realistic precision.

Various sensors are deployed to monitor critical risk points: a wind speed sensor mounted at the top of the crane measures wind speed, ensuring safe operations under adverse weather conditions. Multiple position sensors (#1, #2, #3, #4, #5, and #6) are strategically placed at different locations on the crane to monitor the position and status of mechanical components, ensuring all parts function correctly. A tilt sensor measures the tilt angle of the crane structure, preventing operations beyond safe limits, while a displacement sensor tracks the movement of crane components to ensure precise control during lifting operations. A mobile central unit acts as the data processing hub, collecting and transmitting data from all sensors. Additionally, a video camera provides visual monitoring, enabling remote supervision and quick identification of any operational anomalies.

Together, these sensors and components allow the physical model to simulate real-world operations on-site, collect and transmit real-time data, and provide the virtual model with accurate operational states. This creates a closed-loop system that enhances the safety and efficiency of crane operations.

Virtual Model for Mapping Physical Status

The virtual model is a digital twin of the physical entity, integrating and combining four layers of models: geometric, physical, behavioral, and rule-based. Specifically, the geometric model describes geometric parameters such as dimensions, shapes, and assembly relationships; the physical model analyzes physical properties such as stress, fatigue, and deformation; the behavioral model responds to external drivers and disturbances; and the rule-based model simulates the operational laws and rules of the physical entity, enabling the model to perform functions such as evaluation, optimization, prediction, and assessment, as shown in Fig. 9.

In summary, the virtual model within the tower crane digital twin system is a comprehensive and dynamic simulation tool

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Table 3. Safe status of key points in the climbing process

Overall goal	Decisions	Information requirements		
		Level-3	Level-2	Level-1
Under the premise of ensuring occupation safety, the tower-crane climbing task is completed.	**D1**: Stop unsafe behaviors of operating staff	I1 Projected safety status of operating staff	I1.1 Guidelines for safe behaviors	I1.1.1 Safety technical code for working at height I1.1.2 Safety code for tower cranes
			I1.2 State of health	I1.2.1 Physical health I1.2.2 Mental health
			I1.3 Compliance of behaviors	I1.3.1 Safety helmets I1.3.2 Safety belts I1.3.3 Physical behaviors
	D2: Check safety status and address safety hazards	I1 Projected structural safety status of tower crane	I1.1 Reference for structural safety	I1.1.1 Design rules for tower cranes I1.1.2 Standard for design of steel structures
			I1.2 Force status	I1.2.1 Height I1.2.2 Inclination I1.2.3 Stress of the main chord
		I2 Projected environment safety status	I2.1 Provision of environmental conditions	I2.1.1 Safety technical code for working at height I2.1.2 Safety code for tower cranes
			I2.2 Environment status	I2.2.1 Wind speed I2.2.2 Wind direction I2.2.3 Light intensity
	D3: Confirm the beginning conditions	I1 Projected staff conditions	I1.1 Status of operating staff	—
			I1.2 Status of supervisor	—
		I2 Projected tower-crane conditions	I2.1 Force status	The content is the same as I1.2 in **D2**
			I2.2 Status of key points	I2.2.1 Support between climbing claws and ear plates I2.2.2 Support between forced cross-beam and ear plates I2.2.3 Connections between steel framework and the highest section I2.2.4 Speed of oil jack extending and retracting I2.2.5 Place of new standard section
		I3 Projected environment conditions	I3.1 Environment status	The content is the same as I2.2 in **D2**
	D4: Confirm the performance of operations in each step	I1 Projected sequence and content of operations	I1.1 Reference for standardized tower crane climbing workflow	I1.1.1 Service manual I1.1.2 Tower-crane operating regulations
			I1.2 Status of key points	The content is the same as I2.2 in **D2**
	D5: Intervene in operations	I1 Projected intervention contents	I1.1 Reference for standardized tower crane climbing workflow	—
			I1.2 Status of key points	The content is the same as I2.2 in **D2**

Table 4. Interviewees in GDTA

Role	Background	Experience (years)
Safety expert	Three professors from Huazhong University of Science and Technology	≥10
	Five experts from relevant government departments	≥15
Manufacturer	Nine safety department leaders from Zoomlion Heavy Industry Science and Technology Co. Ltd., XCMG Construction Machinery Co. Ltd. and Yongmao Holdings Ltd.	≥15
Supervisor	Twelve workers from China State Construction Engineering Corporation Ltd. and China Communications Construction Corporation Ltd.	≥10
Operation worker	Thirty workers from China State Construction Engineering Corporation Ltd. and China Communications Construction Corporation Ltd.	≥10

that integrates real-time data from the physical model to deliver accurate, actionable insights into crane operations. This integration enhances safety and optimizes performance, making the virtual model an indispensable component of the digital twin framework.

IoT Based Real-Time Mapping and Dataflow

The internet of things (IoT) system monitors the real-time status of a physical tower crane, playing a crucial role in both sourcing and transferring real-time data, as shown in Fig. 10. The system is structured into several layers: the physical layer, data

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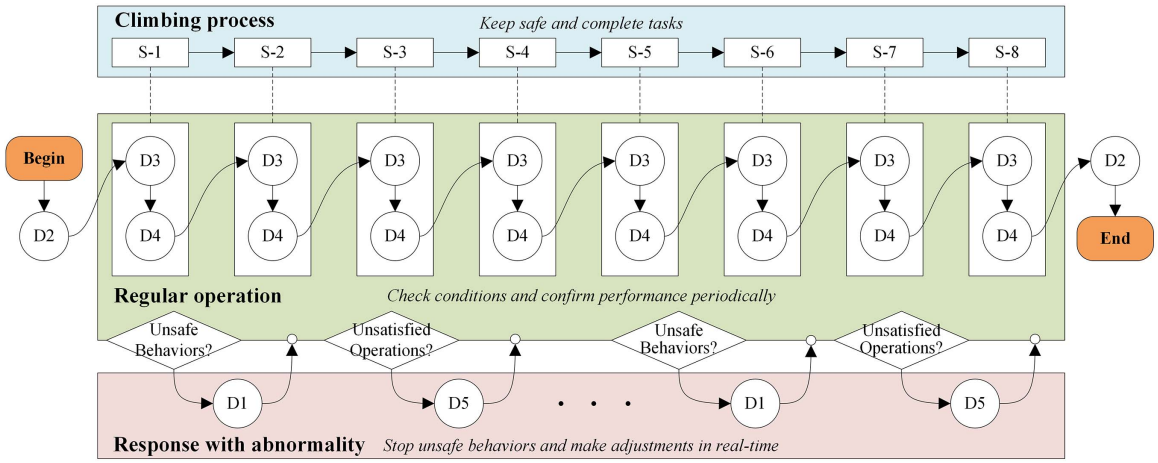


Fig. 7. Distributions of decisions to achieve the major goal during the process.



Fig. 8. Devices deployment in the tower crane model. (Images by Zhang Chong.)

layer, service layer, communication layer, and presentation layer.

Physical Layer: This layer consists of various sensors embedded in or attached to the tower crane, responsible for capturing multisource data such as wind speed, crane inclination, and support status. These sensors monitor critical points essential to the crane's operation.

Data Layer: Sensor data is transmitted via Wi-Fi to an integrated edge intelligent control device (EIND), which processes the multisource data and sends it via 4G to a cloud server. The data layer includes databases that receive, process, and store this data, providing essential information for higher-level processes.

Service Layer: This layer hosts services designed to meet functional requirements such as identifying operational steps, assessing safety risks, and analyzing safety status. These services provide critical information for decision-making and operational guidance.

Communication Layer: The communication layer manages data flow between the front-end and back-end systems using protocols such as hypertext transfer protocol (HTTP), content delivery network (CDN), server load balancer (SLB), Netty, Socket, and HTTP/hypertext transfer protocol over secure socket layer (HTTPS). It ensures seamless real-time data synchronization and processing within the IoT system.

Presentation Layer: The presentation layer consists of user interfaces on mobile terminals and monitoring screens, displaying real-time data visualizations, control options, and safety assessment results. These interfaces allow different users, including operators and supervisors, to access vital information in various scenarios.

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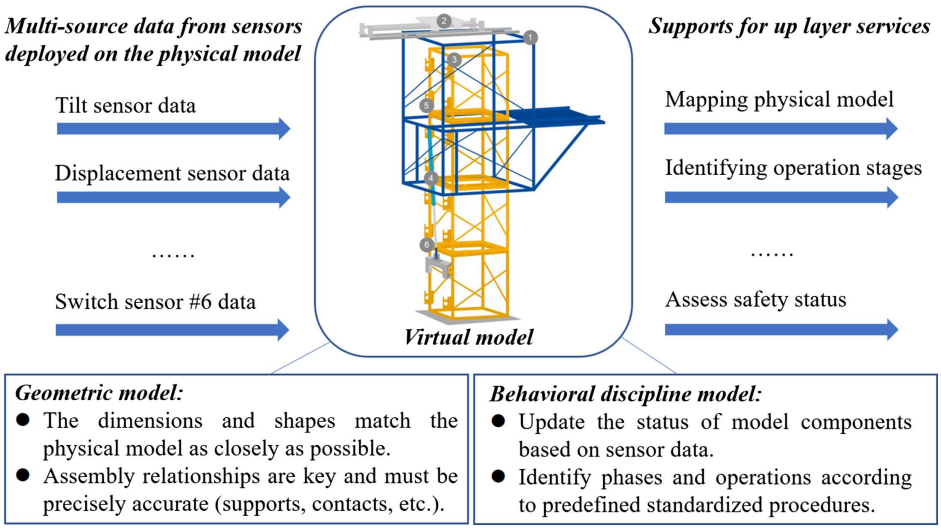


Fig. 9. Structure and discipline of virtual model.

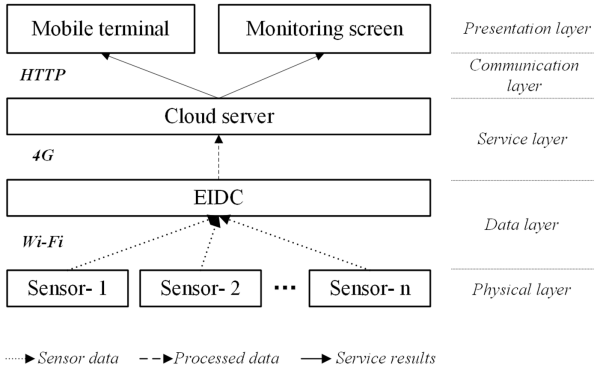


Fig. 10. IoT based data connection.

Real-Time Database for Digital Twin Exchanging

To effectively monitor and manage the real-time status of the tower crane during the climbing process, advanced data integration and processing technologies are essential. This involves seamlessly handling data from various sensors, operational data, and simulation outputs to provide a comprehensive view of the crane's condition throughout the climbing process.

In the tower crane digital twin model, a real-time database is employed to manage this diverse data. The database integrates and processes sensor data from key points such as the cross-beam, climbing claws, tilt angle, and wind speed. This information is critical for assessing safety and guiding operations. The presentation of multisource real-time data is shown in Fig. 11. The duration of this abnormal state has already exceeded the maximum allowed time for the step, thus it transitions to a dangerous status. Specifically, after entering the first climbing phase, the previously free beam must be locked to provide structural support, which requires some time to complete. However, the process of collecting sensor data and determining the beam's status takes less than 1 second. During this time, the operation might still be in progress, and thus cannot be classified as dangerous. However, if the state persists beyond the maximum allowed time, it indicates an unusual issue, and therefore the status transitions to dangerous to trigger an alarm. By leveraging the capabilities of a real-time database, the system efficiently manages data from multiple sources, ensuring accurate and timely updates. This integration enhances both the safety and efficiency of tower crane operations, providing a robust solution for real-time monitoring and management.

Core Service Realization

The workflow for providing safety navigation and real-time alarms is illustrated in Fig. 12. Sensors monitor the status of key points, and the collected data is transmitted to the system platform. Relevant modules use this real-time data to assess the status of these key points, evaluating them against established safety protocols. The results from multiple sources are combined using a Boolean AND operation to generate an overall output. Two judgment criteria—"Is it regular?" and "Abnormal time consumption?"—classify the overall output into three categories: safe, abnormal, or dangerous. For example, in the initial stage, one of the standard procedures is to disconnect the tower body from the lower rotating. Failing to perform this operation may result in the bolts being suddenly ejected when the hydraulic cylinder extends, leading to imbalance of the framework and the potential collapse of the tower crane. From the system's perspective, the initial state will be identified as abnormal, waiting for the operator to promptly disconnect the bolts. If this state persists beyond a certain time threshold, it indicates that the operator likely failed to perform the regular procedure and may proceed to the next step, which requires the system to change the status to dangerous and trigger an alarm. This output is then combined with the safety status of hardhat, determined by a separate module that utilizes a computer vision algorithm, for a final safety classification. The hardhat detection module is developed using the Hikvision AI Open Platform version 2.0 and incorporates an object detection model. An on-site camera equipped with a hardhat detection algorithm transmits workers' hardhat usage status back to the monitoring platform to support upper-level services.

Based on the final safety classification—safe, abnormal, or dangerous—the warning system (including alarms, horns, and terminals) responds accordingly. This ensures that operation staff can follow navigation prompts during climbing procedures or take immediate corrective actions when unsafe conditions are detected.

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Fig. 11. Example of real-time data synchronization. (Images by Zhang Chong.)

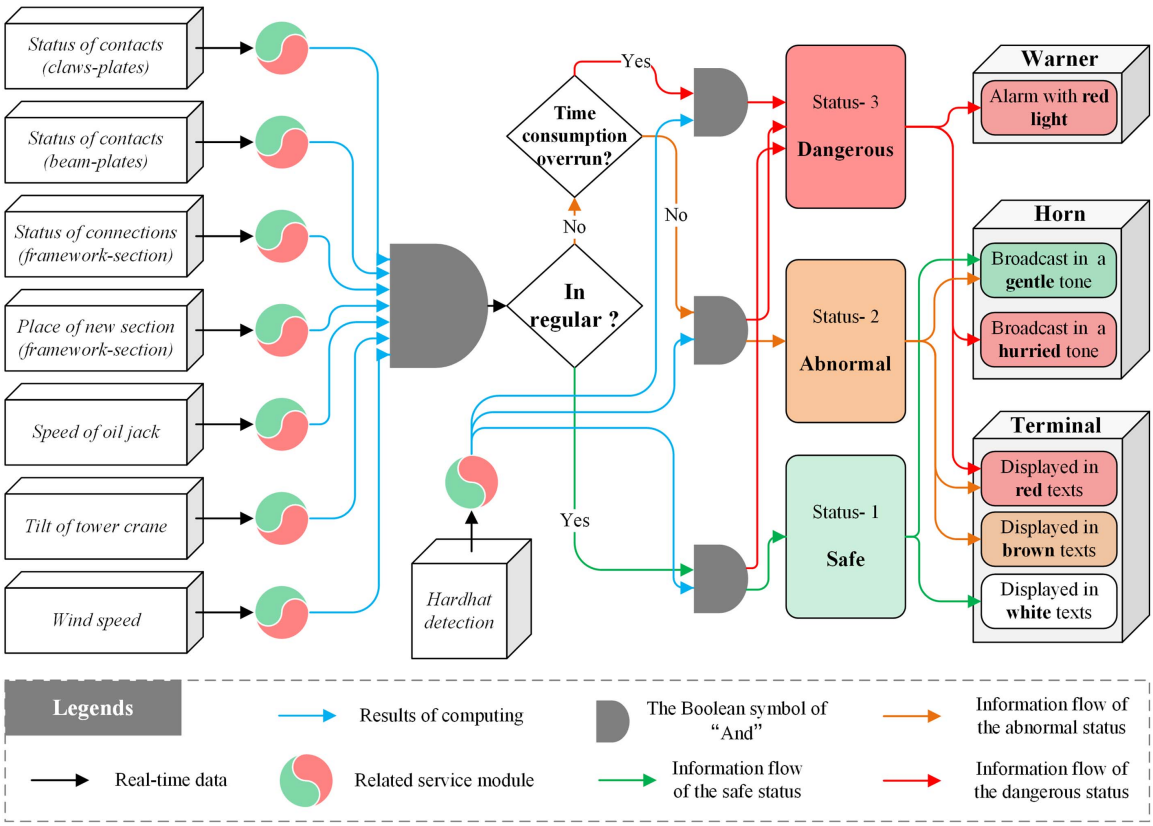


Fig. 12. Workflow of providing navigation and real-time alarm along with the climbing process.

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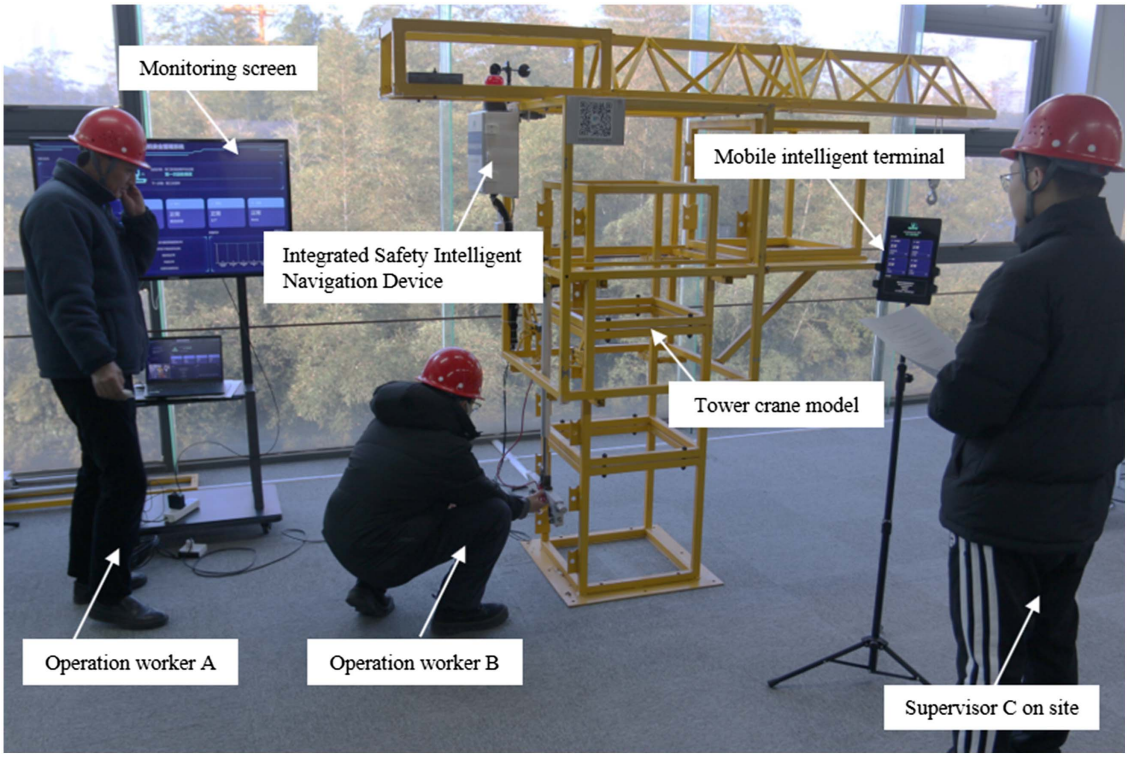


Fig. 13. Experimental platform for the climbing process. (Images by Zhang Chong.)

Practical Effects of the SINP for Ensuring the Process Safety

The performance of the SINP was validated through two phases: indoor functional testing and outdoor practical testing. The indoor validation was conducted at the Key Laboratory of Control Structure in Hubei Province, with technical support provided by Hubei Zhongde Yunxiang Information Technology Group. The outdoor practical validation took place at a construction project in Xiangyang, Hubei, with the assistance of the Hubei Industrial Construction Group.

Indoor Functional Validations

To thoroughly evaluate the proposed method, an experimental platform was established to test the system's performance (Fig. 13). The configuration details of this platform are provided in Table 5.

The first scenario focuses on providing navigation during normal operations, ensuring that all actions are carried out properly throughout the climbing process. The second scenario centers on providing real-time alarms for unsafe conditions, simulating instances of unsafe human behavior and improper operations that would trigger alerts.

Scenario 1: Providing Safety Navigation for Operations

In this scenario, the operation staff completed the tower crane climbing process using two modes:

1. Experience-based operations: The staff made decisions and executed tasks based on their own experience, similar to standard practices on engineering sites.

Table 5. Configuration of the experimental platform

Items in practical	Feature	Items in experimental	Feature
Tower crane	Standard sections	Tower crane model	• $500 \times 500 \times 500$ mm • Three deployed sections and one new section to be added.
	Framework Oil jack Cross-beam Ear planes Climbing claws		• $800 \times 800 \times 1{,}100$ mm • $V_{max} = 0.03$ m/s • All the necessary details for climbing works are replicated in the model. • These parts in the model can effectively work.
Operation staff	First floor	Operation worker A	• Worker A is responsible for adjusting support between forced cross-beam and ear plates.
	Second floor Third floor	Operation worker B	• Worker B is responsible for controlling the oil jack, adjusting support between climbing claws and ear plates, and deploying the new section.
Supervisors on site	On the ground	Supervisor C	• Supervisor C is responsible for supervision of tower crane climbing works.
Auxiliary tool	—	Safety navigation system	• A series of sensors • A safety navigation device • A mobile intelligent terminal • A monitoring screen for the cloud

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Fig. 14. Example of real-time data synchronization. (Images by Zhang Chong.)

2. System-aided operations: The staff followed audio prompts provided by the safety intelligent navigation device (Figs. 14 and 15), which broadcasted navigation instructions that the team followed to complete the tasks.

For comparison, three operational teams were organized to conduct the operations separately: experienced specialized equipment workers, recently trained general workers, and members of the system development team (see Fig. 16). Each team member alternated between the experience-based and system-aided modes to complete a tower crane climbing operation, repeating the process five times.

The key performance indicators recorded during the experiment included the average number of dialogues, average task completion time, and average number of operational errors. The mean values of these metrics are presented in Table 6.

1. Reduction in Communication (Average Number of Dialogues). The average number of dialogues reflects the frequency of verbal communication required during the operation, serving as an indicator of how much coordination or clarification was needed among workers. The significant decline in dialogue frequency in the system-aided mode demonstrates that the system effectively reduced the need for communication. This suggests that the system provided clear and comprehensive real-time information, leading to smoother workflows and less reliance on verbal instructions, thereby enabling more proactive operations.
2. Improvement in Operation Efficiency (Average Task Completion Time). The average task completion time measures operational efficiency. In the system-aided mode, all groups showed a substantial reduction in task completion time. This highlights the system's ability to streamline the operational process, allowing workers to perform tasks more efficiently. The most significant time savings were observed in Group 3 (system development team), indicating that the system may be particularly effective for less experienced users, as it offers real-time guidance and decision support.
3. Elimination of Operational Errors (Average Number of Operational Errors). The average number of operational errors is a critical measure of safety and operational accuracy. In the experience-based mode, all groups recorded operational errors. However, in the system-aided mode, no errors (0) were recorded in any group. The complete elimination of errors in the system-aided mode indicates that the system provides effective guidance, significantly reducing the likelihood of human error and improving safety. This result is particularly noteworthy for Group 3,

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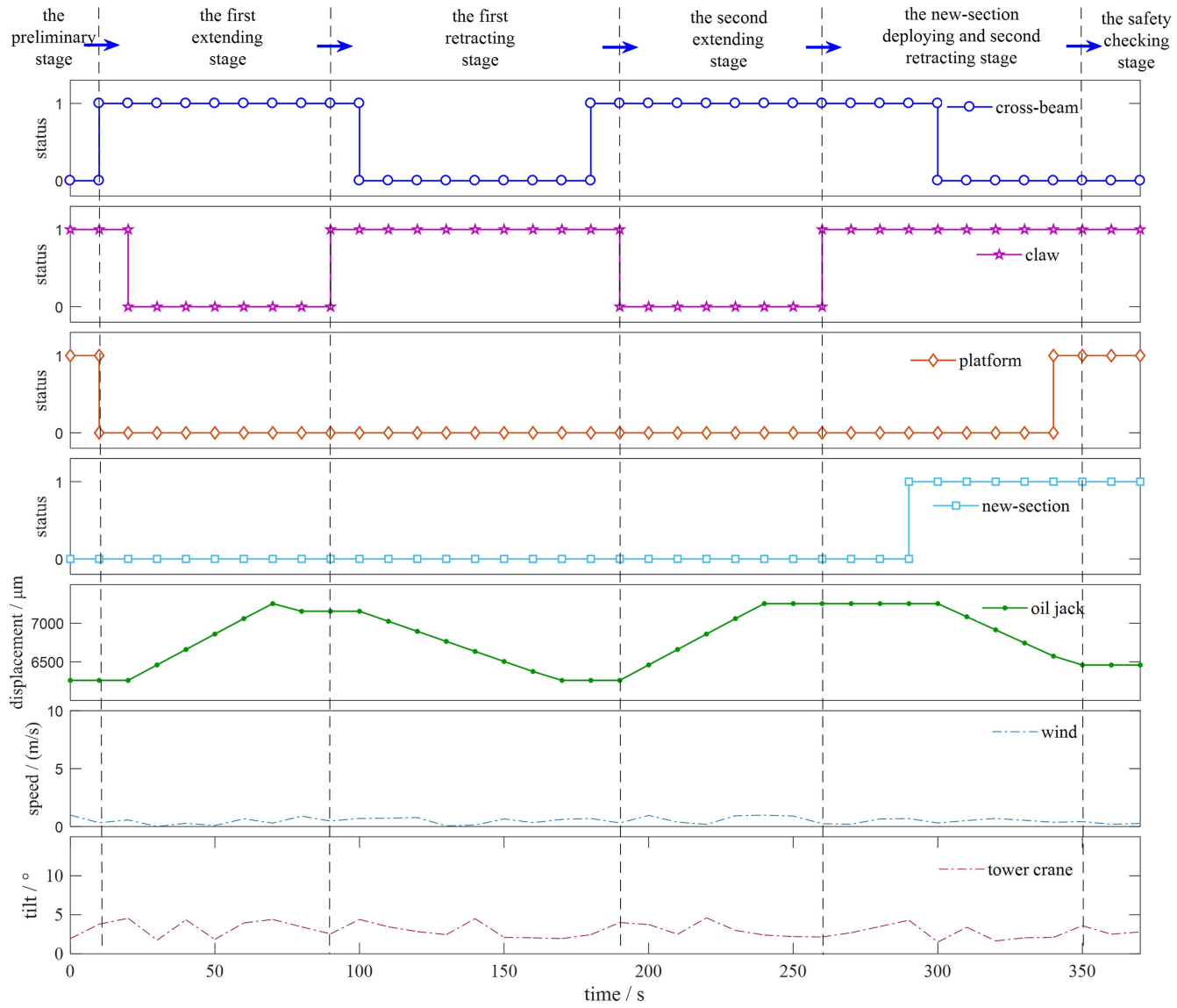


Fig. 15. The status of key points in the first group.

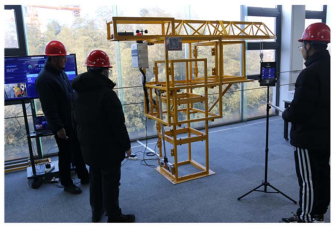
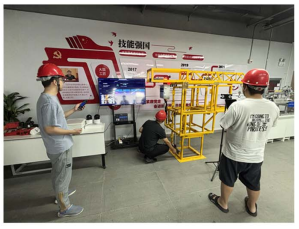
Group	Group 1	Group 2	Group 3
On site			
Crew	Experienced specialized equipment workers	Recently trained general workers	Members of system development team
Feature	They have been involved in tower crane climbing operations for many years, possess a high level of expertise.	They are familiar with tower crane operational procedures, have some practical experience.	They have no practical operational experience, but are very knowledgeable about the system's functions.
Task	Each group members alternately used the experience-based and system-aided modes to complete a tower crane climbing operation, repeating the process 5 times.		

Fig. 16. Staff of three different groups. (Images by Zhang Chong.)

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Table 6. Comparation of performance indicators

Group	Average number of dialogues		Average task completion time		Average number of operational errors	
	Experienced	System-aid	Experienced (s)	System-aid (s)	Experienced	System-aid
Group 1	11.8	0.8	515.6	375.0	1.4	0
Group 2	12	1.6	548.4	444.2	1.8	0
Group 3	12.2	1.0	676.0	391.8	4.8	0

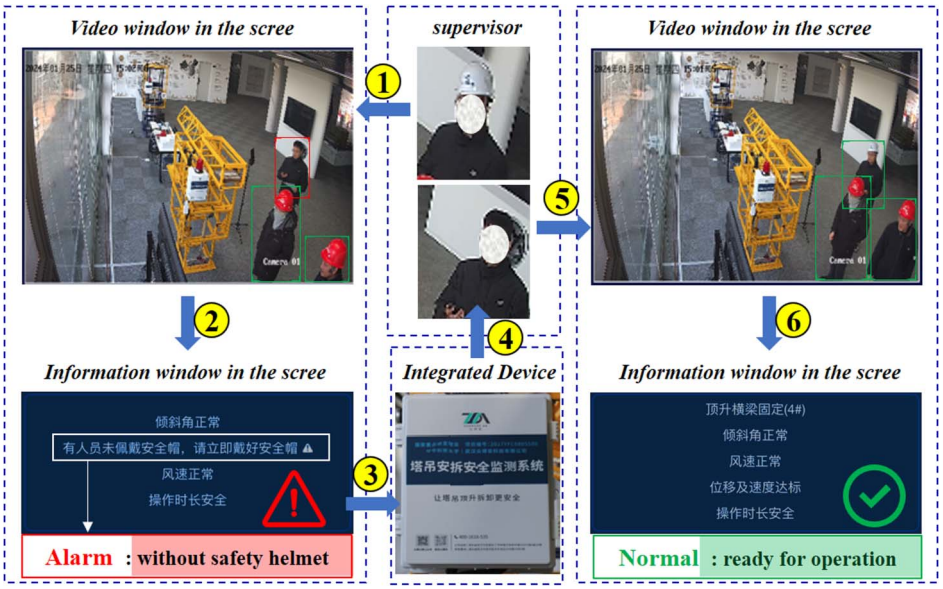


Fig. 17. The working manner of providing real-time alarms for unsafe behaviors. (Images by Zhang Chong.)

which had the highest error rate in the experience-based mode, further emphasizing the system's ability to assist less experienced or trained workers.

Scenario 2: Providing Real-Time Alarm for Unsafe Status

In this scenario, workers deliberately created unsafe situations to trigger the system's alarms. Unsafe behaviors and substandard operations, which are commonly encountered on real-world construction sites, were selected for simulation.

1. Status 1: Unsafe Behavior

   At the start of the climbing process, the supervisor removed his hardhat. The system quickly detected this unsafe behavior using video data and triggered a light and sound alarm: "Somebody isn't wearing a hardhat, please adjust immediately!" The alarm persisted until the hardhat was put back on, as shown in Fig. 17.
2. Status 2: Inadequate Operations

   Three substandard operations were simulated: failing to support the climbing claw with the ear plate, failing to support the cross-beam with the ear plate, and failing to deploy the new section, as shown in Fig. 18.

For example, in Step 3 (see Table 1), the operation staff failed to notice that the climbing claws had not been lifted onto the ear plates. This posed a serious risk, as proceeding to Step 4 without proper support would mean there was no stable force transmission path between the climbing claws and the ear plates. The embedded switch sensor detected this issue and alerted the safety intelligent device. The device recognized the hazardous situation and issued a real-time light and sound alarm: "The climbing claws have not yet been set into position, please check!" The operation staff quickly received the message, made the necessary adjustments, and the alarm was deactivated once the climbing claws were securely positioned. The interrupted process then resumed safely.

The system performed similarly in the other two inadequate operation scenarios: failing to support the cross-beam with the ear plate and failing to deploy the new section.

Outdoor Practical Validation

For the construction project, a series of sensors and an edge intelligent control device (EICD) were installed on a tower crane, as shown in Fig. 19. The sensor configuration on the tower crane closely mirrors that of the experimental model, with certain sensor ranges expanded to meet the tower crane's specifications. Due to project progress requirements, the tower crane needed to undergo a climbing operation. After the operators received a technical briefing and a basic explanation of the system's purpose, they were instructed to collaborate with the navigation system during the operation.

As shown in Fig. 20, the sensors collected real-time physical data from the tower crane, including wind speed, tilt angle, and the status of various components. These perception-level data were transmitted via the IoT to the system for processing. The system aggregated and analyzed the multisource data through the EIDC, producing comprehension-level information. This included assessments of whether unsafe conditions, such as exceeding the safety tilt threshold, existed during the crane climbing process. Simultaneously, the system recognized the current operational phase and performed principle-based analysis to predict upcoming operations

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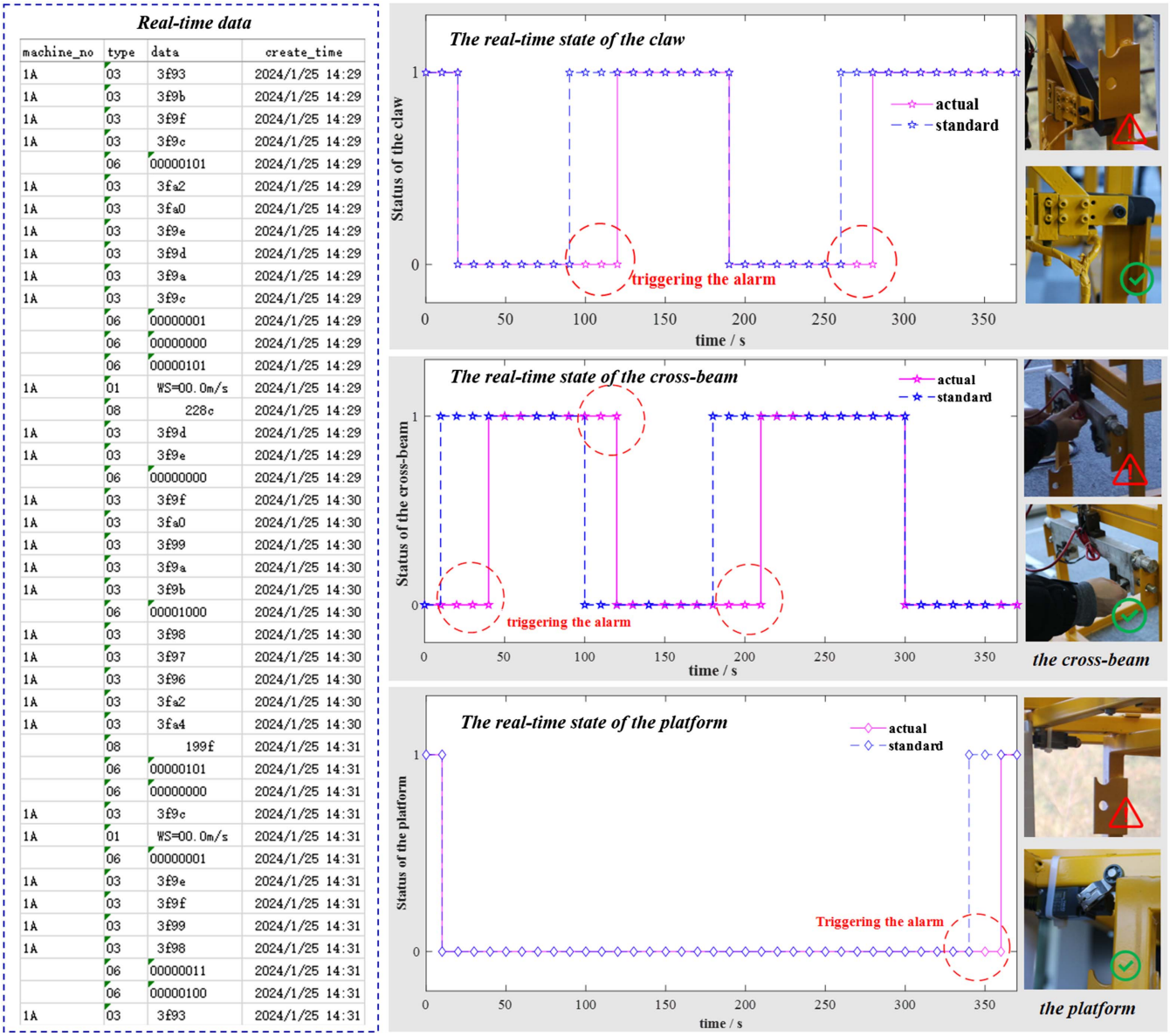


Fig. 18. The working manner of providing a real-time alarm for substandard operation.

and their specific tasks, generating prediction-level information. Based on a comprehensive analysis of all relevant conditions, the system generated prediction-level outputs, such as the crane's overall safety status and safety warnings. These predictive insights were conveyed to nearby operators via audio-visual signals, providing operational guidance and safety alerts. Additionally, the EIDC's analysis results were uploaded to the cloud platform and synchronized with various smart devices, offering intuitive information displays and safety management support for both on-site supervisors and remote leaders.

To assess the practical effectiveness of the system in real-world engineering applications, we designed a user survey targeting 31 individuals involved in tower crane climbing operations, including both operators and supervisors who had used the safety navigation system (see Table 7). The survey compared their experiences with system-assisted tower crane operations against traditional methods, focusing on both operational and supervisory aspects. Eight key indicators were evaluated, including information efficiency, operational efficiency, supervisory convenience, and supervisory effectiveness, with a maximum score of 10 points. The average scores for each indicator are presented in Table 8. The survey results indicate that tower crane operations assisted by the system outperformed traditional methods across all indicators. Notably, the system showed significant advantages in information efficiency (9.1 versus 6.7), process safety (9.5 versus 8.3), supervisory effectiveness (9.1 versus 7.1), and supervisory convenience (9.2 versus 6.9), greatly improving both operational safety and supervisory efficiency. Additionally, the system enhanced operational efficiency and decision-making, with scores of 8.7 and 8.5, respectively. These results further validate the overall effectiveness of the intelligent navigation system in improving tower crane climbing operations.

Discussion

This study presents a novel safety navigation system designed to enhance tower crane climbing operations by improving safety, efficiency, and decision-making through real-time information integration. The findings from the indoor functionality tests, outdoor validation, and user-centered studies demonstrate that the system

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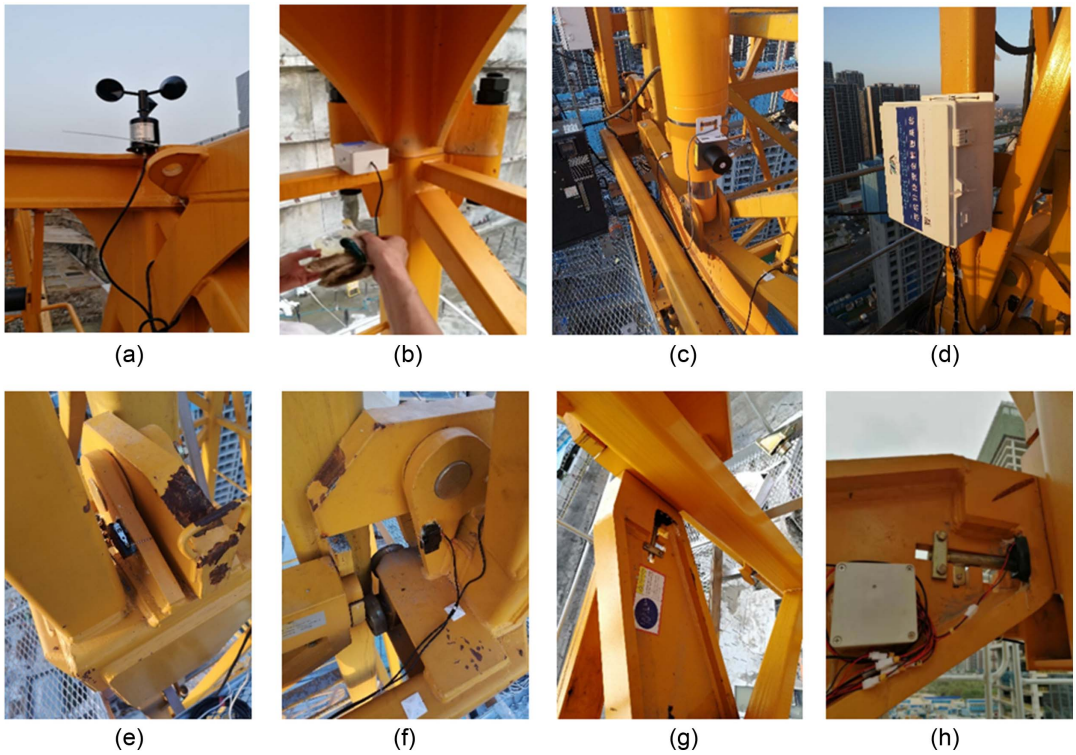


Fig. 19. The deployment of some sensors and EICD: (a) environmental sensor (crane tip); (b) tilt sensor (standard section); (c) displacement sensor (oil cylinder); (d) EICD (steel framework); (e) switch sensor (the left claw); (f) switch sensor (the right claw); (g) switch sensor (beam left side); and (h) switch sensor (beam right side). (Images by Zhang Chong.)

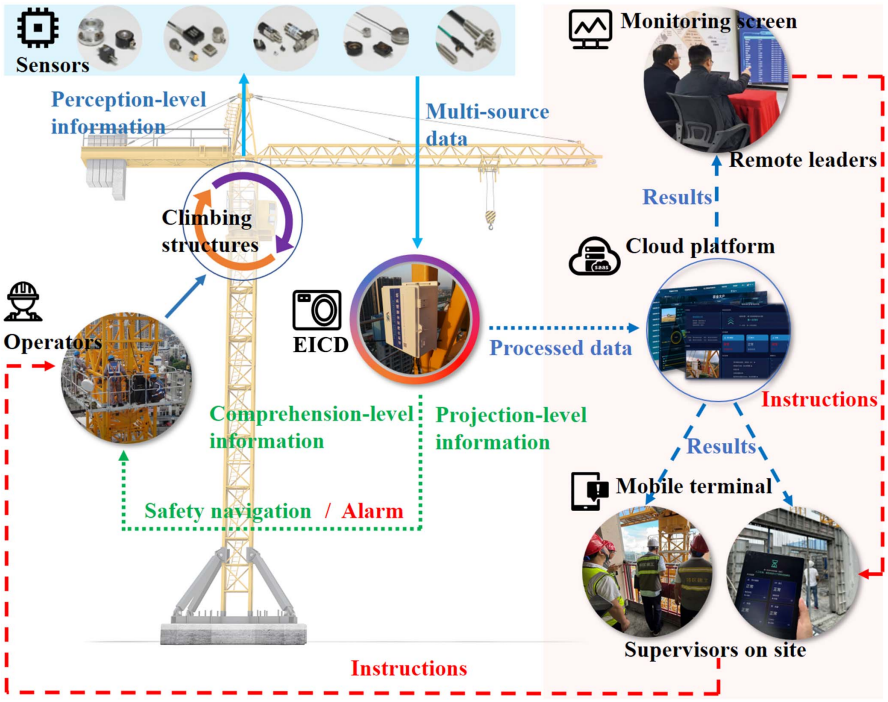


Fig. 20. The implementation of the navigation system in practical projects.

significantly reduces communication needs, improves task completion times, and eliminates operational errors. These results strongly support the system's potential to address critical challenges in tower crane operations.

While the results are promising, several limitations should be acknowledged. First, the study's sample size was limited. This may restrict the generalizability of the findings to broader populations. Second, the system's impact on users' SA was not systematically

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Table 7. The interviewees in user survey

Role	Background	Experience (years)
Safety expert	Three experts from relevant government departments	≥1
Supervisor	Seven workers from China State Construction Engineering Corporation Ltd. and China Communications Construction Corporation Ltd.	≥10
Operator	Twenty-one workers from China State Construction Engineering Corporation Ltd. and China Communications Construction Corporation Ltd.	≥10

Table 8. The average scoring results of tower crane climbing experience

Index	Abbreviation	Experienced	System-aided
Information efficiency	IE	6.7	9.1
Operation efficiency	OE	8.3	8.7
Decision efficiency	DE	7.2	8.5
Process safety	PF	8.3	9.5
Effectiveness of supervision	FS	7.1	9.1
Ease of supervision	ES	6.9	9.2
Reliability of supervision	RS	7.2	8.7
Proactiveness of supervision	PS	6.8	8.6

investigated, leaving room for further exploration of how the system influences decision-making and risk perception in dynamic environments. Last, the testing scenarios, though comprehensive, may not fully capture the diversity of real-world construction contexts.

To address these limitations, three key areas could be executed in depth. First, larger-scale studies with more diverse user groups are necessary to validate the system's effectiveness across various operational contexts. Second, dedicated experiments should be conducted to quantify the system's impact on situational awareness, using established SA assessment methods. Third, applying the system to a broader range of construction scenarios, including high-complexity and high-risk operations, will further test its robustness and adaptability.

Conclusion and Future Works

The tower crane climbing process is inherently high-risk due to its complexity and intricate operations. As demand for higher construction efficiency grows, the frequency of climbing operations has increased, heightening safety concerns. The proposed systematic control framework addresses these risks by guiding workers through safety navigation and providing real-time alarms.

Our research identified several key risk points critical to maintaining tower crane safety and evaluating workers' performance. These include the support between climbing claws and ear plates, the connections between the steel framework and the highest section, and the tilt of the tower crane. Understanding these key points is essential for workers to assess the current status and make informed decisions, offering a conceptual framework for process safety management related to tower cranes.

By developing use case and data sequence models, we established interaction protocols for the components of the SINP to deliver timely safety information. The system's functionality includes providing safety navigation for operations and real-time alarms for unsafe conditions. These capabilities were demonstrated through experiments using a developed system, a simplified tower crane model, and experienced staff. In the indoor functional validation, task performance significantly improved across three groups of participants with different characteristics: the average number of dialogues decreased from 11.8–12.2 to 0.8–1.6, the average task completion time shortened from 515.6–676.0 s to 375.0–444.2 s, and the average number of operational errors dropped to 0 from 1.4–4.8. Additionally, inadequate operations and unsafe behaviors were effectively identified, demonstrating the system's robust capabilities in safety navigation and real-time alarm. In the outdoor practical validation, eight key user-experience indicators, prioritized by 31 users, showed significant improvement: information efficiency (IE) ($6.7 \rightarrow 9.1$), operation efficiency (OE) ($8.3 \rightarrow 8.7$), decision efficiency (DE) ($7.2 \rightarrow 8.5$), process safety (PF) ($8.3 \rightarrow 9.5$), effectiveness of supervision (FS) ($7.1 \rightarrow 9.1$), ease of supervision (ES) ($6.9 \rightarrow 9.2$), reliability of supervision (RS) ($7.2 \rightarrow 8.7$), and proactiveness of supervision (PS) ($6.8 \rightarrow 8.6$). These results indicate that the system provides effective support for industry professionals in safety information sharing, safety navigation, and real-time alarm during tower crane climbing process.

Future research on the systematic framework for tower crane climbing safety should focus on two areas:

1. Integrating structural health monitoring: By incorporating structural health monitoring, the SINP can provide additional safety information related to structural integrity, offering an extra layer of protection during the tower crane climbing process.
2. Incorporating augmented reality (AR): Integrating AR could significantly enhance the effectiveness of navigation and alarms by overlaying real-time operations with standard operational guidelines. This would improve workers' situational awareness and operational precision.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions: Chong Zhang and Wei Zhang conceived the study and were responsible for data collection, analysis and paper writing. Yudong Shi was responsible for practical tests, while Tingsheng Zhao and Yaoting Zhang was responsible for supervision.

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